iGNITION presents

DriveSafe 360

Real-Time Intelligent School Bus Safety System

ESP32/Arduino | AI / ML | Computer Vision | IoT Sensors

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THE PROBLEM

Why Do We Need This?

1.35M

Road fatalities globally every year (WHO)

20%

Of accidents caused by driver drowsiness

50+

Children left behind in hot buses annually

Problem Statement

Design a real-time intelligent bus safety system that monitors driver condition and passenger status, and generates instant alerts to prevent accidents and improve child safety.

OUR SOLUTION

DriveSafe 360 Overview

AI Vision

Webcam-based drowsiness & head pose detection using CNN + MediaPipe face landmarks

Embedded Sensing

ESP32 with ultrasonic, temperature, and alcohol detector

Serial Bridge

Two-way real-time communication between Arduino and AI inference server via USB serial

Instant Alerts

Multi-level buzzer alerts with safety latching for environment hazards until last child exits

The system continuously monitors driver drowsiness/head behavior and bus environment signals, then triggers audible alerts when unsafe conditions are detected.

SYSTEM ARCHITECTURE

How It All Connects

1. Arduino reads sensors (ultrasonic, MQ7, DHT11)
2. Sends telemetry via serial: DIST:xx|COUNT:xx
3. PC AI server processes webcam + serial stream
4. Fusion logic classifies risk level (4 tiers)
5. Alert commands sent back: ALERT_* or OK
6. Buzzer & LCD indicate active risk on bus

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Arduino → Python: distance, count, events, BPM, IMU
Python → Arduino: ALERT_CRITICAL | ALERT_HIGH |
ALERT_MEDIUM | OK
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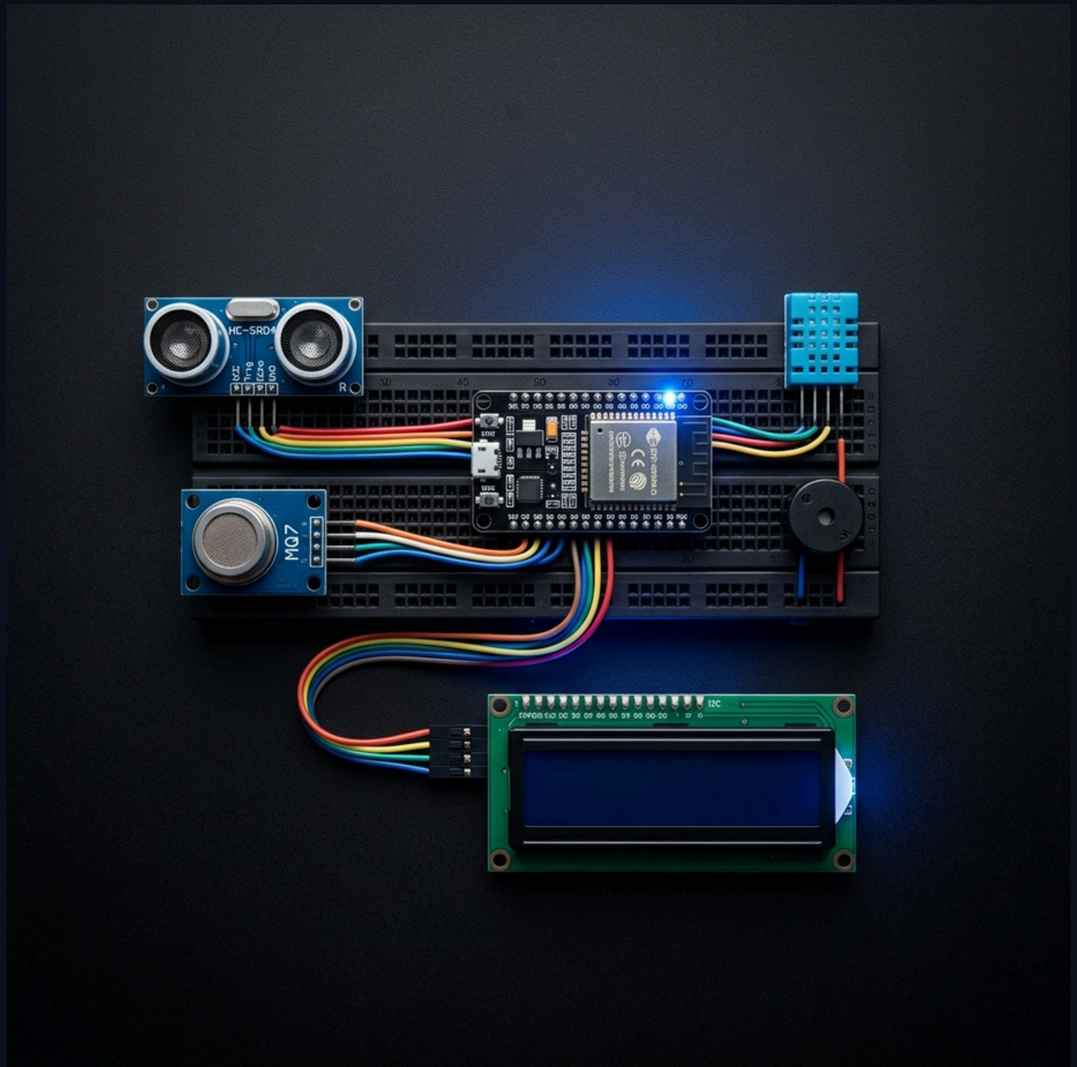
HARDWARE COMPONENTS

The Embedded Layer



MCU	ESP32 DevKit — dual-core, WiFi/BLE capable
SENSOR	HC-SR04 Ultrasonic — passenger counting via trend detection
SENSOR	DHT11 — Temperature & humidity monitoring (alert > 36.5°C)
SENSOR	MQ-7 Gas Sensor — CO/alcohol detection (optional)
OUTPUT	Piezo Buzzer — 4 modes (passive, active HIGH/LOW, dual)
OUTPUT	16×2 I2C LCD — displays count, alerts, sensor data
COMM	USB Serial @ 74880 baud — bidirectional with PC

AI & COMPUTER VISION

Drowsiness & Head Pose Detection

Eye State Model (CNN)

3-layer Conv2D + BatchNorm + MaxPool → Dense(128) → Sigmoid.
48×48 grayscale eye crops, binary open/closed, data augmented.

Head Pose Model (DNN)

Dense(128→64→32) + Dropout → Softmax over 4 classes:
normal, tilt_forward, tilt_side, look_away from yaw/pitch/roll.

PERCLOS + Temporal Fusion

Score = 60% PERCLOS + 25% avg_closed + 15% duration.
Fused closed prob = 70% model + 30% EAR (eye aspect ratio).

MediaPipe Face Landmarks

468 facial landmarks for real-time eye extraction,
EAR calculation, and coarse head pose estimation.

SMART FEATURES

Passenger Counting & Environment Safety

Ultrasonic Trend Detection

- 5-reading median filter removes noise
- Rolling history buffer tracks distance trends
- Monotonically increasing → ENTER detected
- Monotonically decreasing → EXIT detected
- 2-second cooldown prevents double counting
- Occupancy count can never go below zero

Environment Alert Latching

- Temperature threshold: 36.5°C (DHT11)
- Alcohol threshold: MQ7 raw ≥ 1700 (4 confirms)
- Alert latches ON when hazard + children present
- Latch stays active until occupancy = 0
- Prevents clearing alert while kids still on bus
- Buzzer + LCD show active environmental risk

ALERT SYSTEM

Multi-Level Risk Classification

Level	Condition	Action
CRITICAL	IMU detects fall / impact	Immediate continuous buzzer
HIGH	Score ≥ 0.75 + tilt, or eyes closed > 1.2s	Continuous buzzer alert
MEDIUM	Score ≥ 0.55, or eyes closed 0.7–1.2s	Buzzer after 1.2s delay
NORMAL	All metrics within safe range	OK sent, no alert

Smart Suppression: Look-away head state suppresses false drowsiness alerts during mirror checks. Head tilt (forward/side) escalates MEDIUM → HIGH. Face-missing timeout triggers alerts based on last known pose.

PERSONALIZATION

Auto-Calibration & Driver Profiles

Phase 1: Open Baseline

30 seconds of normal driving posture. Captures EAR median and model probability baseline for open eyes.

Phase 2: Blink Baseline

10 seconds of deliberate slow blinks. Calibrates the closed-eye threshold as the midpoint between open & closed medians.

Head Calibration

Computes yaw/pitch/roll center from median values. Adaptive thresholds calculated as 2.6σ + offset, clipped to safe ranges.

Rolling Profiles

JSON files store up to 60 sessions per driver. Active thresholds computed as rolling average of last 10 sessions.

TECH STACK

Technologies Used

Python
Core language for AI server

TensorFlow / Keras
Eye & head model training

OpenCV
Webcam capture & display

MediaPipe
468 face landmarks

scikit-learn
Data preprocessing & splits

PySerial
Arduino ↔ PC communication

ESP32 / Arduino
Embedded sensor platform

I2C / GPIO
LCD & sensor interfaces

FUTURE SCOPE & APPLICATIONS

What's Next

Heart-rate monitoring integration
SMS alert delivery to parents & school
Cloud dashboard for fleet management
GPS route tracking & geofencing
Multi-camera coverage (cabin + exterior)

Applications

- School Buses
- Public Transport
- Fleet Monitoring

TEAM iGNITION

Meet The Team

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Lead — ESP coding, system integration

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Thank You!

Questions? Let's discuss.